2.0T to 3.0T. Moreover, when comparing the experimental results in Figure 6 and Figure 10, we observe a consistent conclusion: as the pre-training data volume increases (from 0.5T to 3.0T), the *rebound* becomes more pronounced. Specifically, when the pre-training data volume increases, the initial performance decline caused by negative data fine-tuning occurs more rapidly, while the subsequent decline slows down. This indicates that larger pre-training data volumes amplify the *rebound* effect in language models.

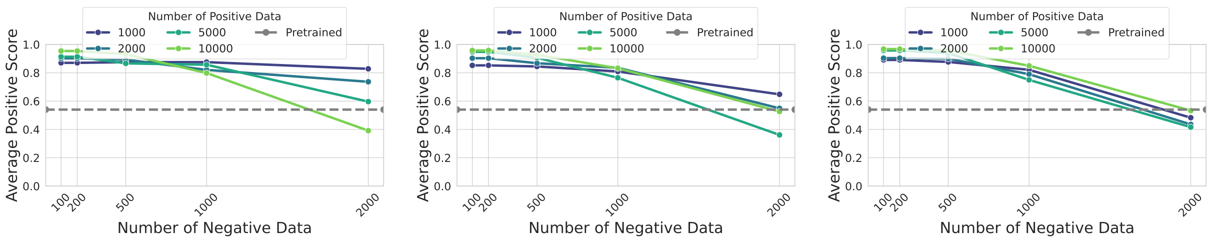

Figure 10: **Experimental results for validating *rebound* increases with model pre-training data volume.** Each subfigure from left to right shows the changes in pre-training data volumes of 0.1T, 0.5T, and 1.0T. As pre-training data volume increases, aligned model performance deteriorates more rapidly after fine-tuning with **negative** data.

C Further Discussions of Language Models' *Elasticity*

C.1 Transfer Learning and Language Models' *Elasticity*

Transfer Learning is formally defined as follows: consider a domain $\mathcal{D} = \{\mathcal{X}, P(X)\}$, which consists of a feature space $\mathcal{X}$ and a marginal probability distribution $P(X)$, where $X = \{x_1, \cdots, x_n\} \in \mathcal{X}$. A task $\mathcal{T}$ within the domain $\mathcal{D}$ is defined by a label space $\mathcal{Y}$ and a predictive function $f(\cdot)$. Given a source domain $\mathcal{D}_S$ and its corresponding task $\mathcal{T}_S$, as well as a target domain $\mathcal{D}_R$ and its task $\mathcal{T}_R$, transfer learning aims to improve the target predictive function $f_R(\cdot)$ by leveraging knowledge from $\mathcal{D}_S$ and $\mathcal{T}_S$, where either $\mathcal{D}_S \neq \mathcal{D}_R$ or $\mathcal{T}_S \neq \mathcal{T}_R$ (Zhuang et al., 2020; Weiss et al., 2016). From the perspective of transfer learning, the pre-training and fine-tuning process of language models can be described as follows: Given labeled training data from a source distribution $\mathcal{D}^s = \{(x_n, y_n) \sim p\}_{n=1}^{N_s}$ and data from a target distribution $\mathcal{D}^t = \{(x_n, y_n) \sim q\}_{n=1}^{N_t}$, we first minimize the empirical risk on the source distribution to fit a model: $f^s = \text{argmin}_f \hat{R}(f, \mathcal{D}^s)$, where $\hat{R}(f, \mathcal{D}^s) = \mathbb{E}_{(x,y)\sim p}[l(y, f(x))]$ represents the empirical risk of model f on the source distribution p. Subsequently, the model is adapted to the target distribution by minimizing, $f^t = \text{argmin}_f \hat{R}(f, \mathcal{D}^t) + \lambda||f - f^s||$, where $||f - f^s||$ measures some distance between the two functions, and $\lambda \geq 0$ is a regularization parameter (Murphy, 2023).

Specifically in the context of language models, transfer learning can almost encompass all models trained under the pre-training-finetuning paradigm. However, transfer learning does not capture the specific details of the model training process, nor does it explicitly incorporate the established theories that explain the varying degrees of difficulty in task transfer for LLMs. Therefore, while transfer learning provides a perspective to analyze various phenomena in LLMs from the standpoint of traditional machine learning, it does not offer concrete tools to explain these *elasticity* phenomena.

Furthermore, in traditional transfer learning settings, positive and negative samples can only be considered as different partitions of the same dataset in classification tasks. However, in our case, these samples are used for autoregressive generation training, where the generation on positive samples and negative samples follows mutually conflicting input-output distributions. This is fundamentally different from classification tasks. Specifically for safe-unsafe generation, steering a language model's generation from an unsafe input-output distribution to the opposite safe input-output distribution is a major challenge in generative language model research. This cannot be adequately explained by merely considering different partitions of the same data distribution.

These considerations lead us to conclude that the observed phenomenon in our work represents a novel behavior that cannot be fully accounted for by existing transfer learning theories. It warrants finer-grained theoretical and experimental investigation. Consequently, we propose the concept of *elasticity* to formally name this new phenomenon observed in language models and introduce corresponding theoretical frameworks and empirical methodologies to examine it. This perspective is empirically supported by the experimental results shown in Table 1. Taking the Alpaca task as an example, we examine the process in which θ_1 and θ_2 are aligned with θ_3. Since θ_2 has acquired more knowledge about the SFT distribution compared to θ_1, it should, from the perspective of transfer learning, find it easier to learn the distribution of θ_3 than θ_1. However, as shown in the experimental results above, this is not supported, suggesting that transfer learning alone cannot fully account for the observed *resistance* phenomenon. [2]

C.2 Discussions of Practical Steps to Mitigate *Inverse Alignment* Risks

The emphasis on *elasticity* phenomenon in language model alignment highlights a crucial challenge for ensuring the safety and robustness of open-source models throughout their entire lifecycle. As our study demonstrates, the resistance of language models to alignment adjustments is fundamentally rooted in the substantial disparity in data volumes across different training phases. This insight suggests a practical mitigation strategy: customizing the scale of synthetic data through the elasticity mechanism offers a feasible pathway for the development of more robust alignment algorithms in the future.

- We discover that the resistance of language models to alignment is essentially due to the significant differences in data volume across various training processes. Therefore, a straightforward idea is to ensure that the training data volumes corresponding to different alignment targets are as similar as possible during the alignment process. This approach helps avoid resistance effects on alignment targets with smaller training data volumes due to subsequent perturbations.

- The elasticity theorem provides a feasible method for customizing data ratios, thereby quantitatively measuring the amount of various types of data needed for a model to meet expected goals. Specifically, for an aligned language model, during subsequent alignment processes, the elasticity theorem indicates that there is an inherent loss of elasticity dependent on data volume ratios when learning unrelated distributions. Therefore, to ensure that previously aligned targets remain satisfied during subsequent alignment, the training objective can be transformed into a constrained optimization problem to quantitatively calculate the data volume required for new target alignment.

- The current challenge in implementing algorithms based on the elasticity theory, mainly lies in the lack of a sufficient theoretical foundation to characterize the specific features of current datasets. This makes it difficult to distinguish the independent and differently distributed premises of different datasets in the elasticity theory. Since real-world datasets often contain fused features, a possible future research direction is to finely characterize the features of datasets with fused characteristics.

C.3 Discussions of Quantitatively Characterize the Impact of Dataset Size on *Elasticity*

Considering that, as indicated in Theorem 4.2, the difference in dataset size is a key factor contributing to the *elasticity* phenomenon, quantitatively characterizing the impact of dataset size on *elasticity* is indeed crucial for both theoretical understanding and practical alignment design. In this section, we present preliminary explorations, based on experimental observations, on quantifying the influence of dataset size on elasticity, as well as the current limitations in achieving this goal.

By synthesizing observations from Figure 6 in the main text and Figure 10 in the appendix, we find that for the tinyllama series models, when the pre-training data volume is merely 0.1T, there is almost no significant resistance phenomenon observed; the positive level of the model's output does not change significantly with an increase in negative data. However, when the pre-training data volume reaches 0.5T, the resistance phenomenon of the language model becomes notably observable. Considering the continuity between pre-training data and model resistance, this suggests that the significant change point of the resistance phenomenon should lie between 0.1T and 0.5T.

Regrettably, there are still challenges in training a series of models with continuously varying pre-training data volumes, and we currently lack sufficiently fine-grained training data samples to conduct such precise series model training. This limitation prevents us from accurately determining the critical

[2]As Reviewer qznt's suggestion, we further discuss the relationship between transfer learning and LLMs' elasticity in this section.

point where the resistance phenomenon becomes significant. Therefore, although we can suggest a possible range, accurately quantifying this threshold in practice remains a considerable challenge. We hope that future research will further address this issue by developing more controllable model series and fine-grained pre-training datasets to enable a clearer, quantitative characterization of *elasticity*.